

UNIT-4

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It is static AC machine which is used for to transfer electrical energy from one circuit to another circuit and its objective is to transfer electrical energy to increase or decrease the value of voltage / current.

The frequency is same for both the circuit

There are two type of transformer.

1. Step-up transformer.

2. Step-down transformer.

(*) Step-up transformer :-

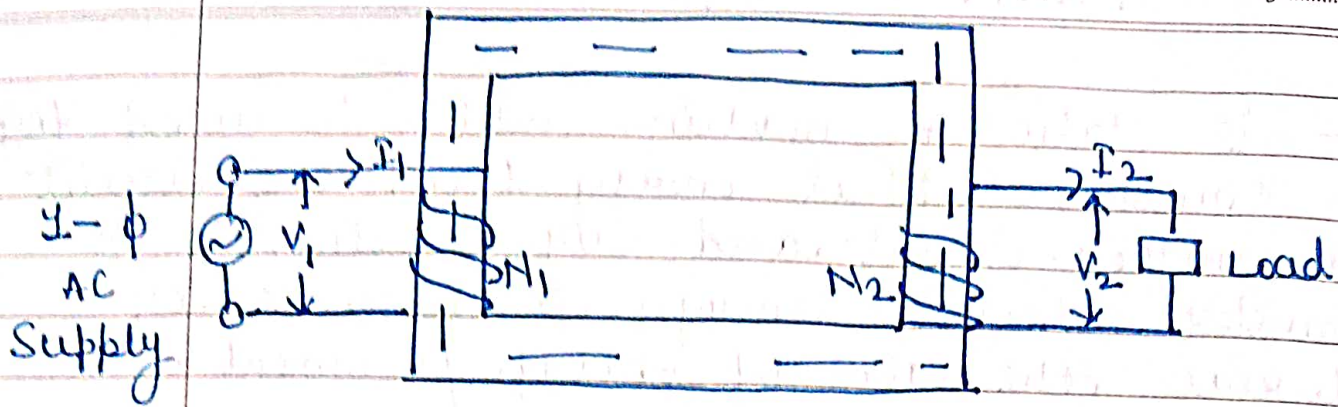
In ~~set~~ step-up transformer low voltage is converted into high voltage. It means the value of $V_1 < V_2$ & $N_1 < N_2$.

*2) step-down transformer:-

In step-down transformer high voltage is converted into low voltage. It means the value of $V_1 > V_2$ & $N_1 > N_2$.

Q. Explain construction & working principle of three phase transformer.

Shreya Mahabadi



Construction:

it consist a core which is made by iron or steel it also consist two winding one is primary winding & another is secondary winding. Supply is given to the primary winding & load is connected to the secondary winding. Magnetic part is core common for both primary as well as secondary winding.

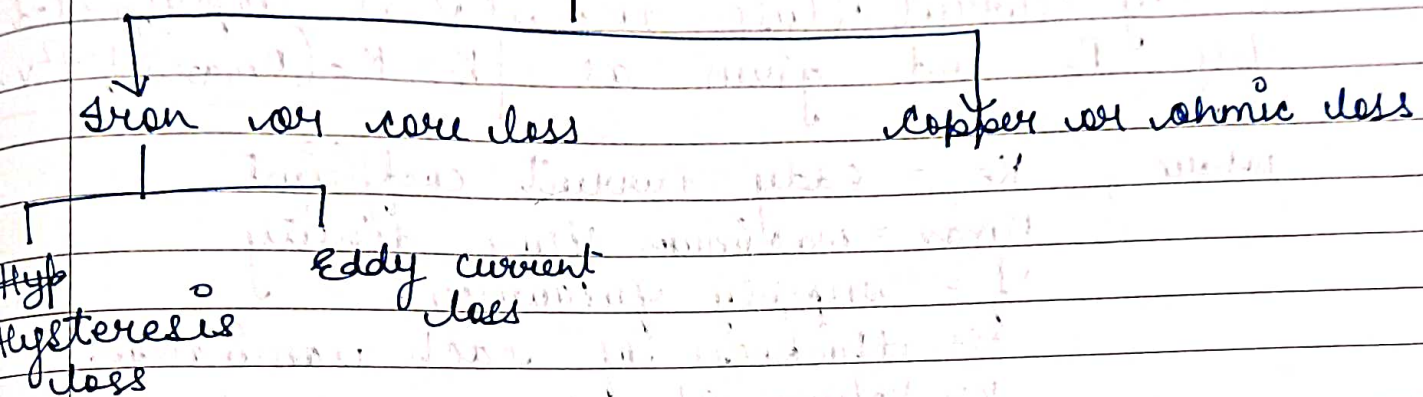
Working principle:

When single phase AC supply is given to the primary winding & alternating current is produced in primary winding. Due to alternating current alternating flux produced in the iron core & as we know from Faraday's law whenever change in ϕ is induced. Secondary winding is also interlinked with iron core therefore ϕ induce on the secondary winding which deliver current across load and this principle

is known as electromagnetic induction principle.

Q. Explain transformer lossage in detail and also derive emf equation in single phase transformer.

⇒ Transformer lossage:



There are two type of transformer lossage

1. Iron or core loss.
2. Copper or ohmic loss.

*1) Iron or core loss: -

It is due to alternating flux in the iron core and there are two type of iron loss:-

1. Hysteresis loss.
2. Eddy current loss.

⇒ 1. Hysteresis loss - is due to magnetising force in the iron core and it is represented by

$$P_h = n' (B_{max})^x \text{ Watt}$$

where n' = Hysteresis coefficient

B_{max} = maximum flux density

depend upon material to material & value is between 1.5 - 2.5 & generally we take 2.5

f = supply frequency.
 V = Volume of core

⇒ 2. Eddy Current loss:- Due to conducting nature of iron some current will flow itself on the surface of the iron core known as Eddy current which is in the form of heating that's why known as Eddy current loss and it is represented by P_e and given as $P_e = K_e (B_{max})^2 f^2 t^2 V$ watt

where,
 K_e = Eddy Current coefficient
 B_{max} = maximum flux density
 f = supply frequency.
 t = thickness of each lamination
 V = Volume of core.

Note:- By reducing thickness of each lamination we can minimise Eddy current loss. & by using high grade steel, we can minimise Hysteresis loss

(*2) Copper loss:-

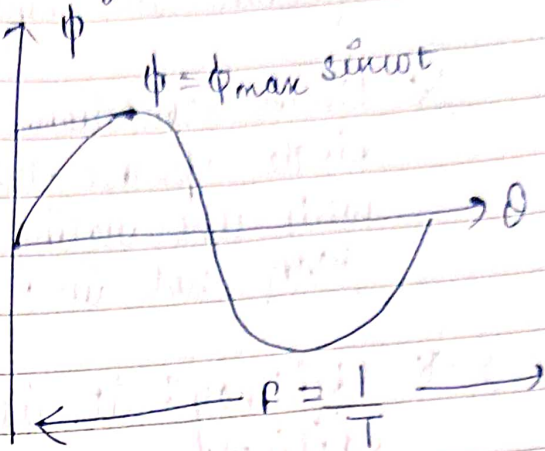
It is due to resistance in primary and secondary winding and represented by $P_c = I_1^2 R_1 + I_2^2 R_2$ watt where,

R_1 = Primary winding resistance
 R_2 = Secondary winding resistance.

Imp

* Derive emf equation in single phase transformer

Let $\phi = \phi_{\max} \sin \omega t$ - (1)
alternating flux produce
in the iron core &
from Faraday law
whenever change in flux
emf is induced which
is given by $E = - \frac{d\phi}{dt}$



$$= - \frac{d(\phi_{\max} \sin \omega t)}{dt}$$

$$= - \phi_{\max} \frac{d(\sin \omega t)}{dt}$$

$$= - \phi_{\max} \omega \cos \omega t$$

$$e = \frac{d}{dt} \phi_{\max} \omega \sin \left(\omega t - \frac{\pi}{2} \right)$$

If the value of emf is maximum then

$$e_{\max} = 2\pi f \phi_{\max}$$

If E_{rms} is the rms of emf then

$$E_{\text{rms}} = \frac{e_{\max}}{\sqrt{2}}$$

$$E = \frac{2 \times 3.14 f \phi_{\max}}{\sqrt{2}}$$

$$E = 4.44 f \phi_{\max}$$

If number of turn is 'N' then

$$E = 4.44 N f \phi_{\max}$$

This is required expression for emf equation in single phase transformer.

Q. Derive condition for maximum efficiency in single phase transformer and also explain why the rating of transformer is in KVA not in KW.

⇒ * Efficiency: it is represented by η & defined as ratio of output power to the input power.

$$\eta = \frac{\text{output power}}{\text{input power}} = \frac{O/P}{O/P + \text{Losses}}$$

$$= \frac{O/P}{O/P + P_i + P_c}$$

* Derivation for maximum efficiency :-

Let

V_2 = output voltage

I_2 = ^{output} Input voltage current

$\cos \phi$ = power factor

P_i = iron loss

P_c = copper loss

x = load at which efficiency will be maximum

$$\eta = \frac{V_2 I_2 \cos \phi}{V_2 I_2 \cos \phi + P_i + P_c}$$

$$\eta = \frac{x P}{x P + P_i + x^2 P_c} \quad \text{--- (1)}$$

Diff eq ① w.r.t x

$$\frac{d\eta}{dx} = \frac{(xP + P_i + x^2 P_c)P - xP(P + 2xP_c)}{(xP + P_i + x^2 P_c)^2}$$

$$\frac{d\eta}{dx} = \frac{P(xP + P_i + x^2 P_c - xP - 2x^2 P_c)}{(xP + P_i + x^2 P_c)^2}$$

$$\frac{d\eta}{dx} = \frac{P\{P_i - x^2 P_c\}}{(xP + P_i + x^2 P_c)^2}$$

for maximum efficiency $\frac{d\eta}{dx} = 0$

$$\Rightarrow \frac{P\{P_i - x^2 P_c\}}{(xP + P_i + x^2 P_c)^2} = 0$$

$$\Rightarrow P\{P_i - x^2 P_c\} = 0$$

$$P \neq 0$$

$$\therefore P_i - x^2 P_c = 0$$

$$\Rightarrow x^2 P_c = P_i \quad (\text{variable loss})$$

when copper loss is equal to iron loss (constant loss) then the value of efficiency will be maximum.

* In Transformer iron loss depend upon voltage and copper loss depend upon current both losses do not depend upon power factor that is why the rating of transformer

is in KVA and not KW.

$$V_1 I_1 = V_2 I_2$$

Primary rating is equal to secondary rating.

$$\frac{V_1}{V_2} = \frac{I_2}{I_1} = \frac{N_1}{N_2} = K$$

where 'K' is transformation ratio

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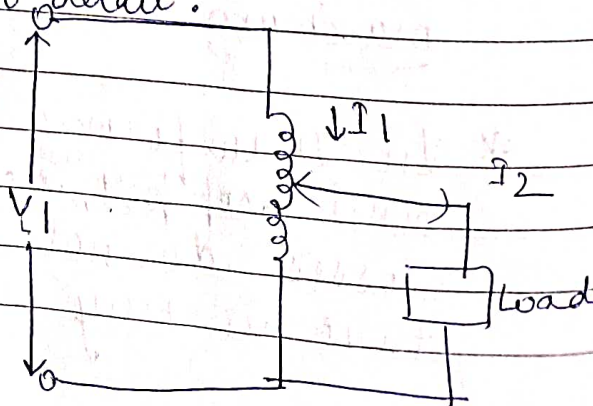
Q:

Why transformer is not used for AC supply? When we provide AC supply to the primary winding a current ^{will} flow which is constant in magnitude ~~that is~~ the value of flux will also be constant and from Faraday law the value of emf induced is equal to zero. It means there is not ~~any~~ secondary current and excessive high amount of current flow through primary winding which may be damage or burn out the transformer. i.e., transformer ^{is} never used in AC supply.

Q:

Explain Autotransformer in detail:

⇒ In autotransformer, primary and secondary winding are inter related to each other & electric & magnetic part is common for both the



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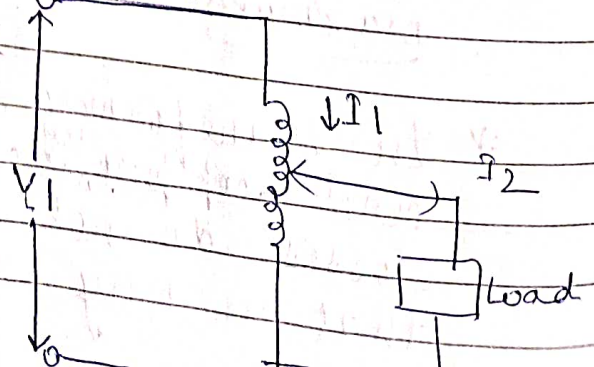
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Q: Explain Autotransformer in detail:

⇒ In autotransformer, primary and secondary winding are inter related to each other & electric & magnetic part is common for both the



winding. By varying number of turns we can vary the value of voltage. In Autotransformer less material required compared to ordinary transformer. Therefore, it is more economical and cheaper than ordinary transformer.

Application:

1. It is used as starter.
2. Where variable voltage is required.

Measurement:

Measurement is a process in which we compare an unknown value with known value. In electrical engineering we measure current with the help of ammeter, voltage with the help of voltmeter, power with help of wattmeter and energy with the help of energy meter and ^{resistance} with help of ohm meter.

* Measuring instrument:

With the help of measuring instrument we perform measurement. There are two type of measuring instrument:-

1. Absolute type.
2. Secondary type
 - Indicating → deflection of pointer
 - Recording → recorder measurement for short duration
 - Integrating → measurement for long duration.

* Torque acting on any electrical measuring instrument:

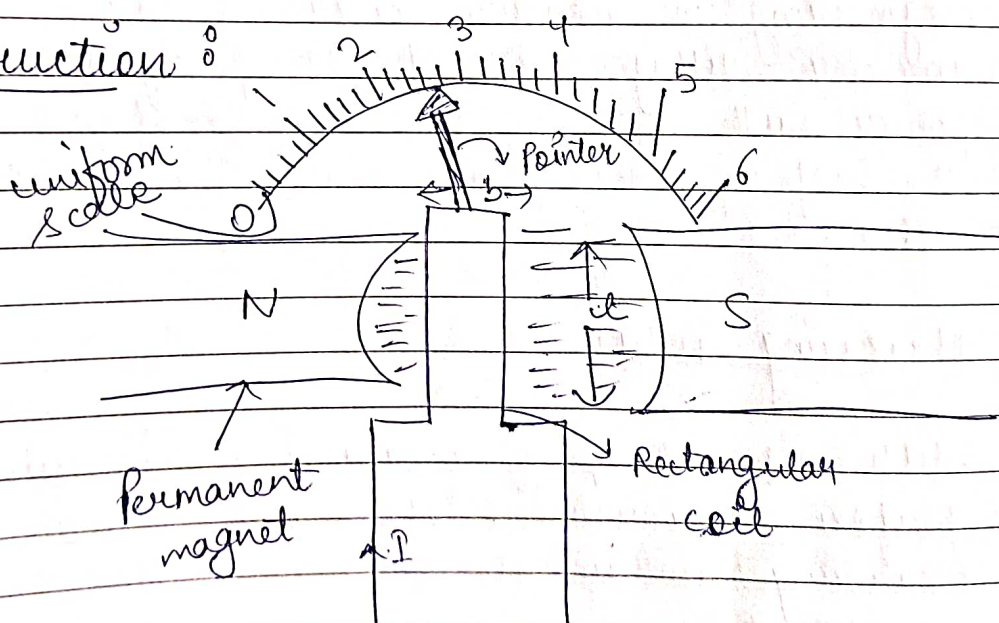
There are three torque acting on any AC electrical measuring instrument:

1. Deflecting Torque
2. Damping torque
3. Controlling torque.

V.V Imp

Q. Explain construction & working principle of permanent magnet moving coil instrument. (PMMC)
(used for measuring current or voltage)

⇒ Construction:



It consists a permanent magnet which is alloy of ALNICO (Aluminium, nickel & cobalt) rectangular coil, pointer, uniform scale & spring type instrument.

Working Principle:

When a current carrying conductor is placed in a magnetic field it experience a mechanical force due to this mechanical force a deflecting torque is produced which is responsible for deflection of pointer.

* Derivation for deflecting torque.

Let,

l = length of coil

b = breadth of coil.

A = area of coil.

N = number of turn.

θ = deflection of pointer.

I = current supply to the coil.

T_c = controlling torque

T_d = deflecting torque.

B = magnetic field strength.

When supply current given to the rectangular coil and placed in a magnetic field it experience a mechanical force which is given by

$$F = IBl \sin \theta$$

Due to this mechanical force a deflecting torque is produced which is given by

$$T_d = F \times b$$

$$= IBl \sin \theta \times b$$

$$T_d = IBA$$

If N is number of turn

$$T_d = NIBA$$

If B, A, N are constant then

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$$T_d \propto I$$

In PMMC we used spring control type instrument for controlling torque and controlling torque is directly proportional to θ .
At equilibrium

$$T_c \propto T_d$$

Therefore, $\theta \propto I$

~~Where~~

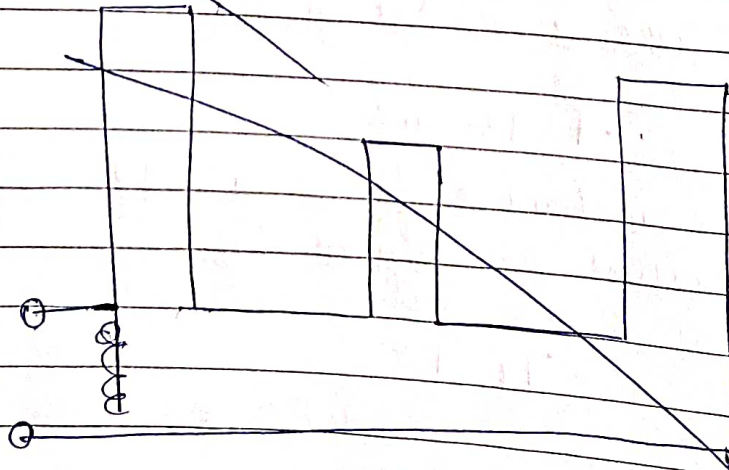
⇒ Advantages :

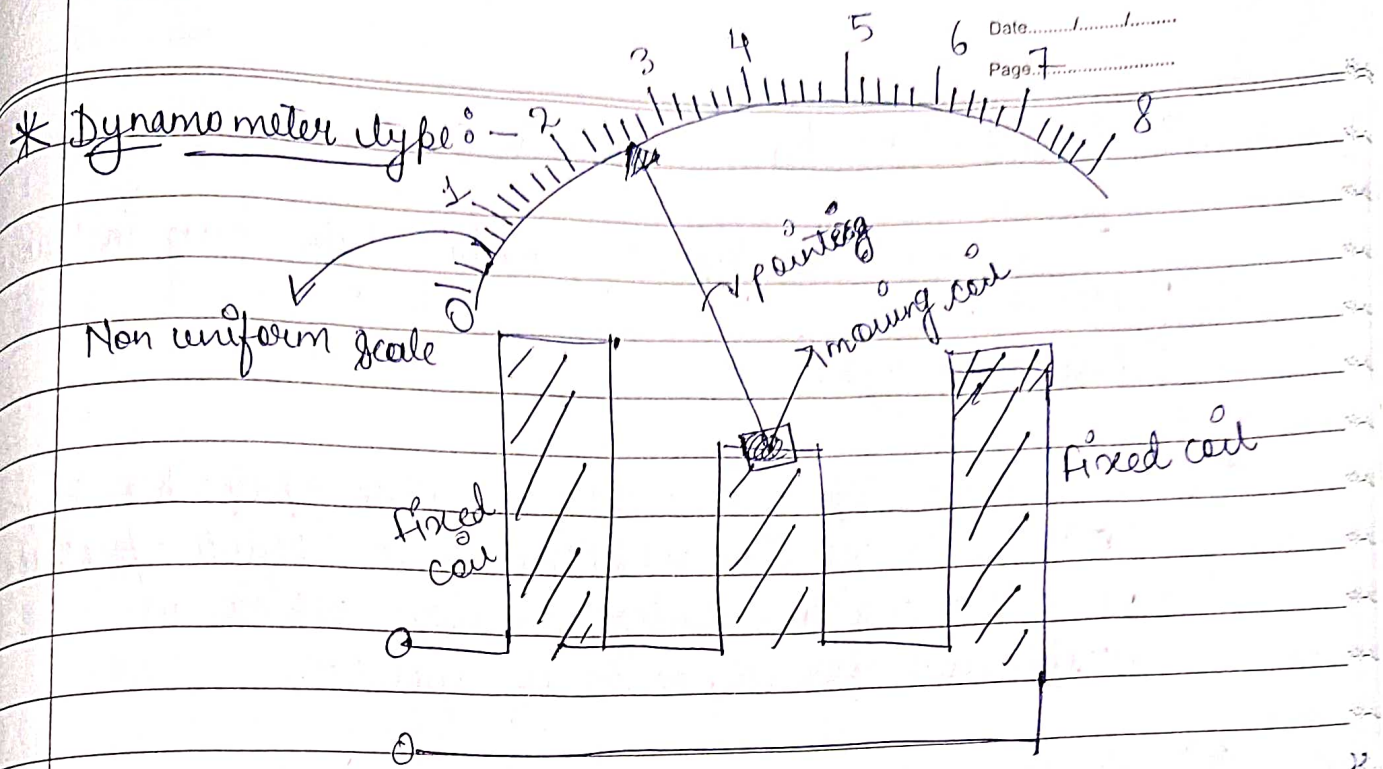
1. Low power consumption.
2. More reliable.
3. Minimum hysteresis and eddy current loss.
4. Torque weight ratio is high.
5. Uniform scale.

⇒ Disadvantages :

1. Only used for DC measurement.
2. Due to delicate construction it is so costly.

* Dynamo meter type :-





In Dynamometer type instrument, two coils are used one is fix coil which is divided into two parts and another is moving coil which is attached to the pointer.

By varying the magnetic field of fix coil we produce two different types of magnetic field which is responsible for deflection of pointer.

If I_p = Fixed coil current
 I_m = moving coil current
 $I_p = I_m = i$
 & as we know
 $T_d \propto i^2$

Advantages —

1. It works on both AC & DC
2. Simple in construction.
3. More economical.

Disadvantages —

1. Non-uniform scale.

* Moving Iron type Instrument -

There are two types of moving type iron instrument -

(1) Attraction type.

(2) Repulsion type.

* (1) Attraction type - it works on the principle of attraction of two opposite field which produce a force which develop a torque which is responsible for deflection of pointer.

* (2) Repulsion type - it works on the principle of repulsion of two opposite & similar field which produce a force which develop a torque which is responsible for

* Formula for Numerical :

1. Load corresponding maximum efficiency $\Rightarrow$

$$x = \text{Rated KVA} \times \sqrt{\frac{P_i}{P_c}}$$

2. Efficiency $\Rightarrow \eta = \frac{x V_2 I_2 \cos \phi}{x V_1 I_2 \cos \phi + P_i + x^2 P_c} \times 100$

3. At full load $\Rightarrow x = 1$

4. At half load $\Rightarrow x = \frac{1}{2}$

5. Maximum efficiency $\Rightarrow \eta_{\max} = \frac{V_2 I_2}{V_2 I_2 + 2 P_i \text{ or } 2 P_c} \times 100$

0.671
0.819

25000

$\frac{10^5}{102340}$

48.85

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$$c = \frac{10}{5} c' = 6 \frac{10}{2}$$

Q. A transformer is rated at 100 KVA at full load its copper loss is 1400 watt and iron loss is 940 watt. Calculate (1) the efficiency at full load, unity power factor.

(2) Efficiency at half load, the same power factor.
(3) the load KVA at which maximum efficiency will occur.

$$\Rightarrow (1) \quad \eta = \frac{1 \times 1400 \times 940 \times 1}{1 \times 100 \times 10^3 + 1400 + 1^2 \times 940} \times 100$$

$$= \underline{97.71\%}$$

$$(2) \quad \eta = \frac{\frac{1}{2} \times 10^5 \times 1}{\frac{1}{2} \times 10^5 + 1400 + \left(\frac{1}{2}\right)^2 \times 940} \times 100$$

$$= \underline{97.4\%}$$

$$(3) \quad 100 \times \sqrt{\frac{1400}{940}} \quad 100 \times \sqrt{\frac{940}{1400}}$$

$$\Rightarrow \underline{81.9 \text{ KVA}}$$

Extension of range of PMMC.

(1) - Ammeter shunt

$$R_{sh} = \frac{R_m}{\frac{I}{I_m} - 1}$$

S_h = Shunt resistance
 R_m = Ammeter resistance
 I = Current to be measured
 I_m = Full scale deflection current

(2) Voltage multiplier

$$R = \frac{V}{I_m} - R_v$$

V = Voltage to be measured
 I_m = full scale deflection current
 $R_v = R_m$ = Volt meter resistance

Q. A moving iron instrument having an internal resistance of 50Ω indicates full scale deflection with a current of $40mA$. How can we make it to work. (1) Voltmeter to read $100V$ on full scale
(2) Ammeter of $4A$ on full scale.

$\Rightarrow$ (1) $V = 100$
 $I_m = 40mA = 40 \times 10^{-3} = 10^{-2} A$
 $R_v = 50\Omega$

$$R = \frac{100}{10^{-2}} - 50\Omega$$
$$= 10^5 - 50 = 9950\Omega$$

(2) $I = 4A$
 $I_m = 10^{-2} A$
 $R_m = 50\Omega$

$$R_{sh} = \frac{50}{\frac{4}{10^{-2}} - 1} = \frac{50}{100 - 1} = \frac{50}{99} = 0.505\Omega$$